

L4 Dealing with data sets

Name: _____

Class: _____

Date: _____

Time: **56 minutes**

Marks: **47 marks**

Comments:

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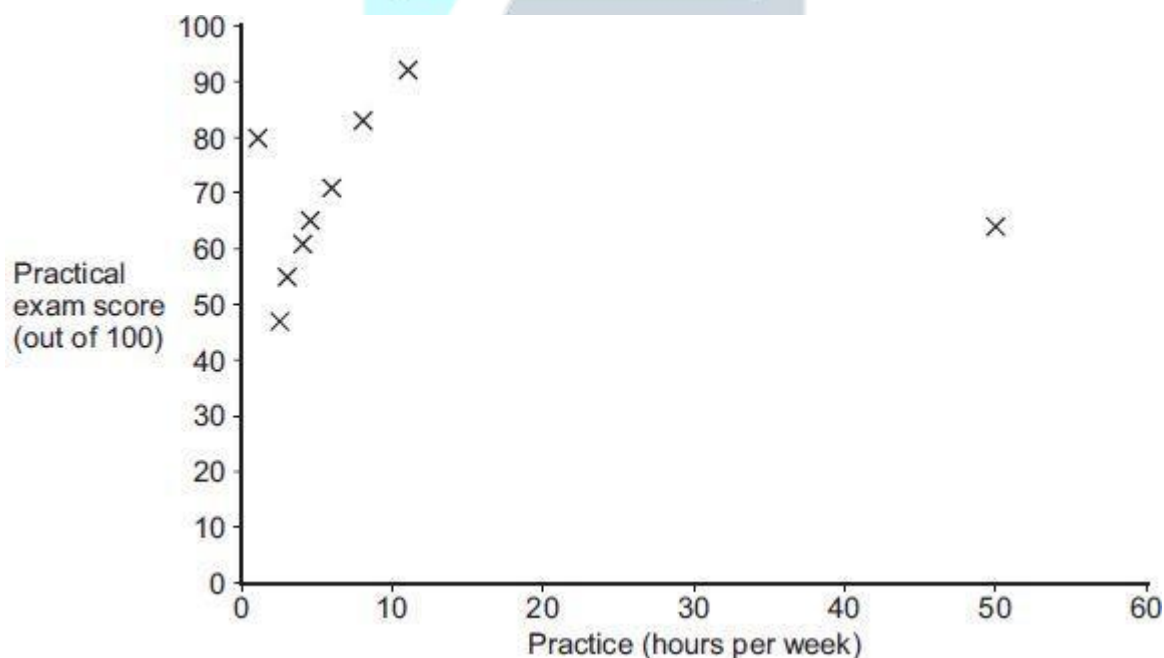
Q1.

Jennie is a piano teacher who teaches nine pupils.

She records how many hours per week they practice the piano along with their most recent practical exam score.

Student	Practice (hours per week)	Practical exam score (out of 100)
Donovan	50	64
Vazquez	6	71
Higgins	3	55
Begum	2.5	47
Collins	1	80
Coldbridge	4	61
Nedbalek	4.5	65
Carter	8	83
White	11	92

She plots a scatter diagram of this data, as shown below.



- (a) Identify two possible outliers by name, giving a possible explanation for the position on the scatter diagram of each outlier.

(4)

(b) Jennie discards the two outliers.

(i) Describe the correlation shown by the scatter diagram for the remaining points.

(1)

(ii) Interpret this correlation in the context of the question.

(1)

(Total 6 marks)

Q2.

The table contains an extract from the Large Data Set.

	Units	2005-06	2007	2009	2011
Confectionery	g	122	126	131	130
Chocolate bars - solid	g	31	31	30	31
Chocolate bars - filled	g	53	55	58	56
Chewing gum	g	2	3	2	2
Mints and boiled sweets	g	33	35	37	37
Mints	g	4	4	4	3
Boiled sweets	g	28	30	33	34
Fudges, toffees, caramels	g	4	3	4	3
Takeaway confectionery	g	0	0	0	0

(a) Bilal states that there is an error in the Large Data Set because the figures for Mints and boiled sweets in the 2007 column do not total to 35.

Give a reason why Bilal's statement may be incorrect.

(1)

(b) Maria claims that there is no need to collect Takeaway confectionery data because the table shows that nobody purchases any of that category of confectionery.

State, with a reason, whether you agree or disagree with Maria's claim. Use your knowledge of the Large Data Set to support your answer.

(1)

(Total 2 marks)

Q3.

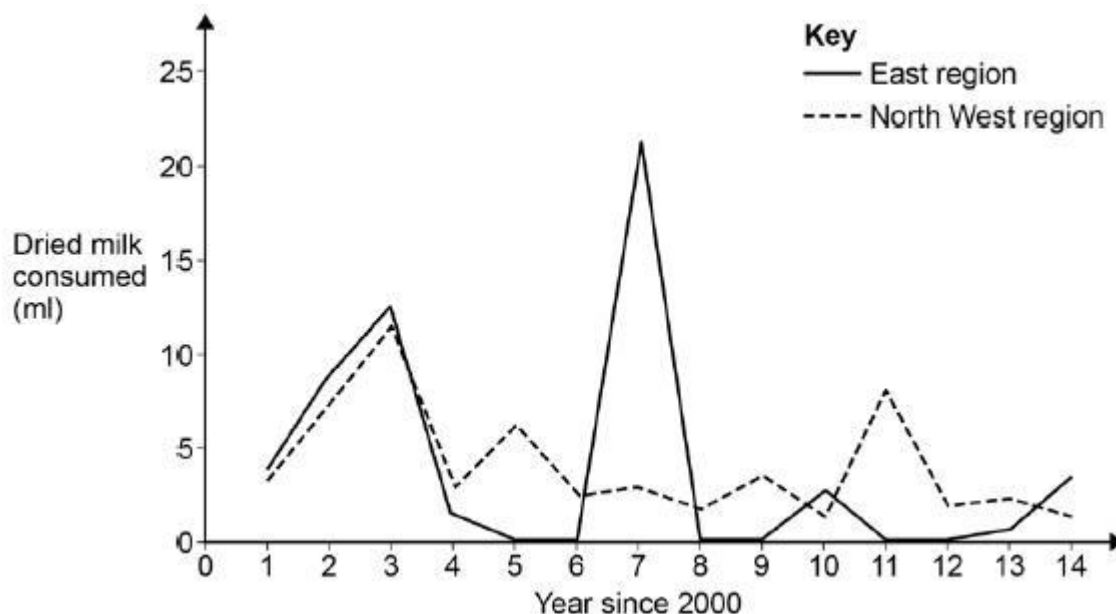
The data in the table gives the average amount, in millilitres, consumed per person per week of dried milk products for the East (E) and for the North West (NW) regions.

The line graph illustrates these data.

Year since 2000													
1	2	3	4	5	6	7	8	9	10	11	12	13	14

E	4	9	12	1	0	0	21	0	0	3	0	0	1	0
NW	3	7	12	3	6	2	3	2	3	1	8	2	2	1

Line graph to illustrate average dried milk consumption for E and NW regions



- (a) (i) Identify, by year and region, a likely outlier. (1)
- (ii) A value in a data set can be identified as an outlier if it exceeds $Q_3 + 1.5 \times IQR$ where IQR is the Interquartile range. (3)
- Finding any necessary statistics from the given data, confirm that the suggested outlier in part (a)(i) can be confirmed as an outlier.
- (b) For the region with the confirmed outlier, explain why the mean is **not** an appropriate measure of average to use. (1)
- (c) Suggest a reason for the occurrence of the identified outlier. (1)
- (Total 6 marks)

Q4.

Some information from the Large Data Set is given in Figures 1 and 2 below.

Figure 1

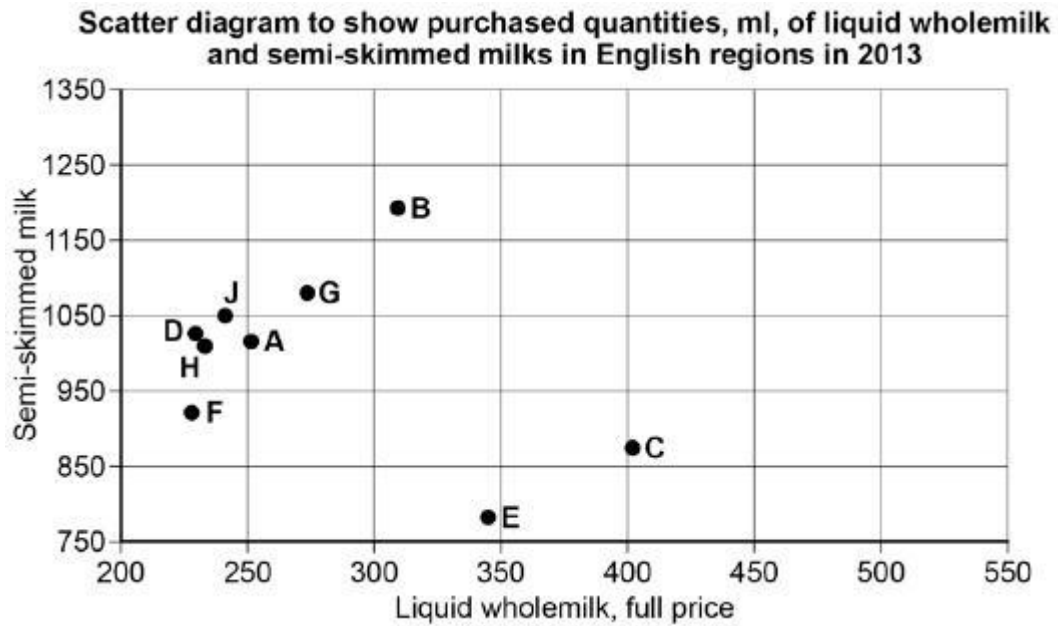
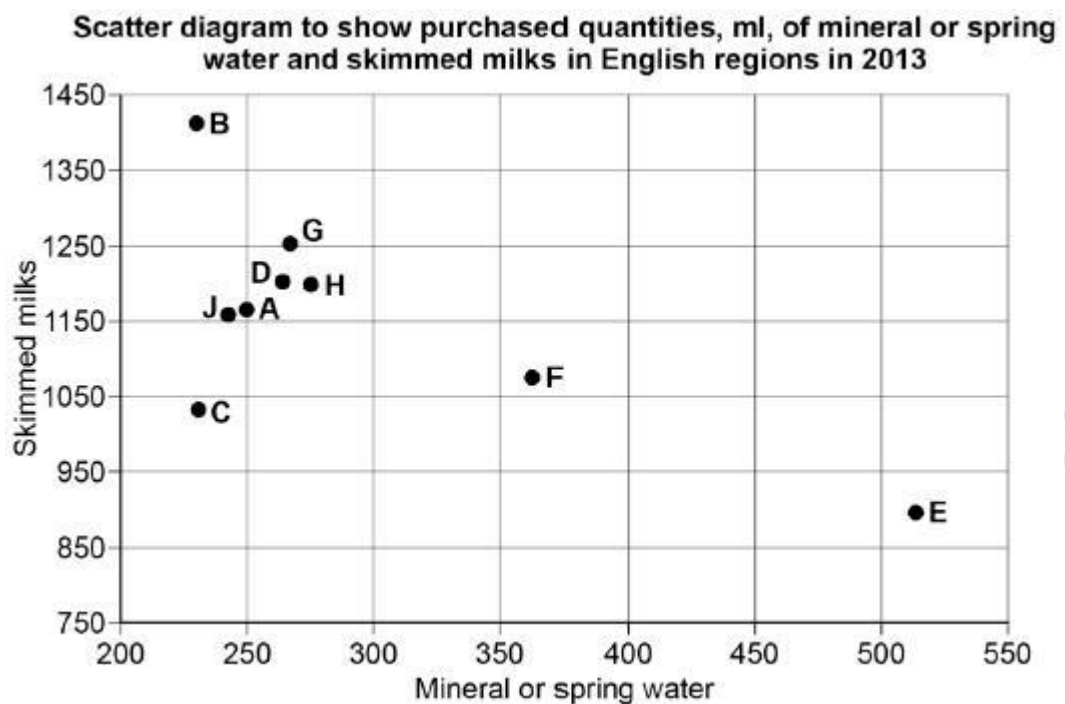


Figure 2



- (a) Give a reason why the recorded vertical data values are higher for each region in Figure 2 than in Figure 1
- (1)
- (b) (i) Describe the correlation between 'Semi-skimmed milk' and 'Liquid wholemilk, full price'.
- (2)
- (ii) Bilal claims that Figure 2 indicates that when people drink more mineral or spring water they tend to drink less skimmed milk.

Comment on Bilal's claim.

(2)

- (c) Suggest, with a reason, which region is indicated by the letter E. Use your knowledge of the Large Data Set to support your answer.

(2)

(Total 7 marks)

Q5.

The number of matches in each of a sample of 85 boxes is summarised in the table.

Number of matches	Number of boxes
Less than 239	1
239–243	1
244–246	2
247	3
248	4
249	6
250	10
251	13
252	16
253	20
254	5
255–259	3
More than 259	1
Total	85

- (a) For these data:

- (i) state the modal value;

(1)

- (ii) determine values for the median and the interquartile range.

(3)

- (b) Given that, on investigation, the 2 extreme values in the above table are 227 and 271, calculate estimates of the mean and the standard deviation.

(4)

- (c) For the numbers of matches in the 85 boxes, suggest, with a reason, the most appropriate measure of spread.

(2)

(Total 10 marks)

Q6.

The number of matches in each of a sample of 85 boxes is summarised in the table.

Number of matches	Number of boxes
Less than 239	1
239–243	1
244–246	2
247	3
248	4
249	6
250	10
251	13
252	16
253	20
254	5
255–259	3
More than 259	1
Total	85

(a) For these data:

(i) state the modal value;

(1)

(ii) determine values for the median and the interquartile range.

(3)

(b) Given that, on investigation, the 2 extreme values in the above table are 227 and 271:

(i) calculate the range;

(1)

(ii) calculate estimates of the mean and the standard deviation.

(4)

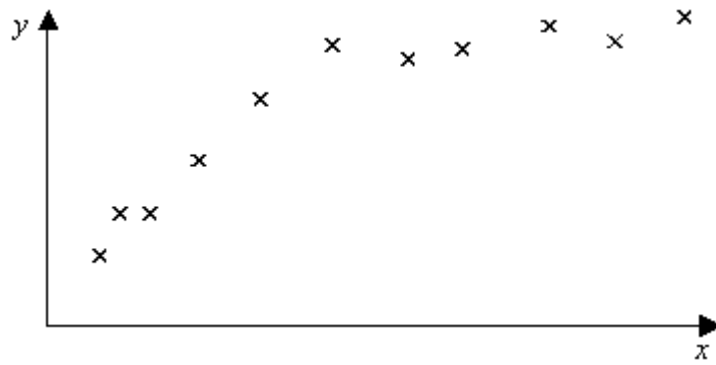
(c) For the numbers of matches in the 85 boxes, suggest, with a reason, the most appropriate measure of spread.

(2)

(Total 11 marks)

Q7.

The scatter diagram below shows, for a sample of countries, average annual income per head, x , and the percentage of children in full-time education, y .



- (a) Describe the relationship between x and y .

(3)

- (b) Comment on the suggestion that the strength of the relationship should be measured by calculating the product moment correlation coefficient between x and y .

(2)

(Total 5 marks)



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